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EPS : An Explainable Post-Shot Expected Goal Metric for Evaluating Goalkeepers and Attackers

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Abstract

We present **EPS** (*Explainable Post-Shot*), a calibrated and intrinsically interpretable model of the probability that an *on-target* shot results in a goal in soccer. Unlike black-box post-shot models that require retrofitted explanations, EPS is a Generalized Additive Model with pairwise interactions (GA^2M) trained via Explainable Boosting Machines (EBMs). Its univariate and bivariate shape functions provide direct, human-readable mappings from execution variables (such as shooter and goalkeeper positions, ball endpoint, or speed) to goal probability. The model's probabilistic outputs are natively well calibrated, a property we establish theoretically and confirm empirically. Building on per-shot predictions, we introduce recruitment-oriented indices. The **Goalkeeper Rating** (GKR) measures the normalized differential between total EPS faced and goals conceded. For attackers, two finishing metrics, $\mathbf{STR}^1$ and $\mathbf{STR}^2$, quantify the ability to convert pre-shot xG into post-shot danger and to translate danger into goals. Finally, we propose the **Transfer Metric** (TRAM), which projects finishing performance across leagues using k-nearest neighbours on shot descriptors. Together, these components yield a transparent, statistically principled framework that supports scouting, recruitment, and performance evaluation.

Introduction

A single goal often decides a football match. Quantifying how a shot becomes a goal, and distributing credit between striker and goalkeeper, is therefore central to modern performance analysis and recruitment. The pre-shot notion of *expected goals* (xG) estimates scoring probability at the moment of the attempt based on contextual factors such as distance, angle, or defensive pressure. Post-shot models (often labeled PSxG or xGOT) instead condition on the fact that the ball is on target and evaluate the likelihood of scoring given its projected crossing point, speed, and height. While the latter provide more insight into finishing quality and goalkeeper performance, many implementations remain black-box, relying on ensembles or neural networks whose predictions must then be "explained" retroactively.

We present **EPS** (*Explainable Post-Shot*), a metric designed to be both accurate and intrinsically interpretable. EPS is built upon Explainable Boosting Machines (EBMs), a modern extension of Generalized Additive Models: each feature and selected interaction is represented by a smooth or piecewise-constant shape function mapping execution variables (shooter location, goalkeeper position, ball endpoint, speed) directly to log-odds of scoring. This construction yields human-readable effects: analysts can verify, for instance, that shots at mid-height near the posts or high-speed strikes from distance shift probabilities in intuitive ways.

A key property of EPS is that predicted probabilities are *well calibrated*. Whereas generic ensembles often require isotonic or Platt scaling, EBMs tend to align with empirical frequencies because they optimize a proper scoring rule (log-loss) with shallow boosted trees, restrict interactions to low-dimensional terms, and employ shrinkage/bagging to limit variance. Empirically, the aggregate number of goals predicted by EPS

closely matches the number observed in our test set, and reliability diagrams confirm that predicted probabilities track observed frequencies across bins. Calibration, here, is not an afterthought but a consequence of the model class and training protocol.

Beyond single-shot predictions, EPS underpins evaluation metrics relevant to clubs and recruitment. The Goalkeeper Rating (GKR) contrasts the total danger faced with actual goals conceded. For attackers, two indices (STR^1 and STR^2) quantify the ability to transform pre-shot xG into post-shot danger and then into goals. Finally, a Transfer Metric (TRAM) leverages k-nearest neighbours on shot descriptors to project how a player's finishing would translate across leagues.

EPS therefore addresses three shortcomings of current practice: the opacity of black-box post-shot models, the unreliability of poorly calibrated outputs, and the lack of consistent, recruitment-oriented indices. It is simultaneously interpretable, statistically principled, and directly applicable to scouting and performance evaluation.

1 Related Work

The rise of expected goals (xG) has established probabilistic modeling as a central tool in sports analytics. Early formulations in ice hockey relied on logistic regressions with distance and angle [Macdonald, 2012], and analogous ideas transferred to football as event and tracking data matured. Subsequent work enriched pre-shot models with spatiotemporal context, possession chains, and pressure, notably through valuing actions [Decroos et al., 2019] and threat models such as Expected Threat (xT) [Singh & Cervone, 2018]. These approaches are invaluable for chance creation and tactical evaluation, but they do not isolate the execution of the strike or the goalkeeper's contribution once the ball is headed on target.

Post-shot models (PSxG/xGOT [Stats Perform, 2020]) explicitly condition on the ball being on target and typically rely on tree ensembles or neural networks. While effective, these methods are opaque and require *post-hoc explanation techniques* such as LIME [Ribeiro et al., 2016], SHAP [Lundberg & Lee, 2017], or Integrated Gradients [Sundararajan et al., 2017]. Such tools approximate feature attributions but can be unstable, sensitive to distribution shifts, and difficult to communicate to practitioners [Slack et al., 2020]. In contrast, Generalized Additive Models (GAMs) strike a balance between interpretability and flexibility by representing covariate effects as smooth functions [Hastie & Tibshirani, 1990, Wood, 2017]. Explainable Boosting Machines (EBMs) extend this framework by learning shape functions with boosted shallow trees and selectively adding pairwise interactions, often matching black-box accuracy while retaining glass-box structure [Lou et al., 2013, Caruana et al., 2015, Nori et al., 2019]. This combination is particularly suited to sports settings where stakeholders require both performance and intelligibility.

Calibration is equally crucial. Miscalibrated probabilities distort evaluations: an attacker may appear clinical merely because the model is over-confident; a goalkeeper may appear weak if save likelihoods are systematically underestimated. The literature recommends assessing both discrimination and reliability, with post-hoc fixes such as isotonic regression or Platt scaling when necessary [Zadrozny & Elkan, 2002, Platt, 1999, Niculescu-Mizil & Caruana, 2005, Naeini et al., 2015, Guo et al., 2017]. In our setting, the EBM's structure and training already yield strong calibration, reducing the need for corrective mapping.

Finally, projecting performance across leagues can be approached with non-parametric analogies. Classic k-nearest neighbours [Cover & Hart, 1967] compare a player's actions to historical counterparts in a target competition. Embedding such instance-based projections within a calibrated post-shot framework enables more credible transfer estimates for finishing and goalkeeping—questions that lie at the heart of recruitment.

2 Problem Statement & Methods Framework

2.1 Data and Split Strategy

We analyze *non-penalty* and *non-deflected* shots from the first and second divisions of France, Italy, Belgium, Spain, Portugal, and England over the 2023/24 and 2024/25 seasons. In line with our evaluation protocol, the 2024/25 season data for Portugal and England are excluded from model training and reserved for external assessment. The resulting corpus comprises **18 081** on-target *non-goals* (saved or blocked) and **7 464** goals,

corresponding to a goal ratio of **29.2%** among on-target attempts. Data are collected by FootoVision[1].

For each attempt we record, at strike time, the shooter location ($\texttt{startX}, \texttt{startY}$), the goalkeeper location ($\texttt{gkX}, \texttt{gkY}$), a body-part indicator $\texttt{header} \in \{0, 1\}$, and the ball speed $\texttt{speedKmh}$. To prevent outcome leakage we compute leakage-safe goal-plane intercepts ($\widehat{\texttt{endY}}, \widehat{\texttt{endZ}}$) from the initial shot direction (see Sec. 2.3), rather than from any save location. We also experimented with richer descriptors such as shot angle and the number of defenders inside the shooter–posts cone. On stratified validation splits, these features produced no material improvement (AUC and ECE differences remained within sampling noise), while degrading model transparency. They are therefore excluded from the final specification.

Splitting is a critical aspect for valid inference on finishing and shot-stopping: distributional shifts across leagues and goalkeeper pools can otherwise induce spurious skill signals. Following concerns raised by [Davis & Robberechts, 2024] about biases in expected-goals evaluations, we adopt a stratified hold-out strategy balancing three dimensions: (i) league tier (major vs. secondary European leagues), (ii) season segment, and (iii) the *goalkeeper population*. The latter criterion is designed to approximate the *average professional goalkeeper*: we control the distribution of shots faced across keepers so that the validation split is neither dominated by a handful of elite shot-stoppers nor by over-exposed keepers from struggling teams. This prevents the learner from overfitting to the presence of strong keepers in training and supports external validity when applying EPS across leagues with heterogeneous goalkeeper levels.

2.2 Set-up and target of inference

We consider a set of random variables defined on a common probability space. At the time of each shot attempt, we observe covariates

$$X = (X_1, \ldots, X_M) \in \mathcal{X} \subset \mathbb{R}^M,$$

together with the binary outcome $Y \in \{0, 1\}$ indicating whether the attempt results in a goal, and an indicator $T \in \{0, 1\}$ specifying whether the shot is on target (i.e., between the posts).

Our object of interest is the *conditional post–shot success probability*

$$p^\star(x) = \mathbb{P}(Y = 1 \mid X = x,\, T = 1)\,, \qquad x \in \mathcal{X}, \tag{1}$$

which represents the chance that a shot with covariates x becomes a goal, conditional on being on target. In practice, $p^\star(x)$ is unknown and must be approximated. We therefore introduce a transparent statistical model and denote its prediction by $\widehat{p}(x)$.

Definition (EPS). We define the *Explainable Post–Shot (EPS)* probability as the model prediction

$$\mathrm{EPS}(x) := \widehat{p}(x),$$

that is, a model-based approximation of the target quantity $p^\star(x)$. For a specific attempt i with covariates x_i, we write $\mathrm{EPS}_i = \mathrm{EPS}(x_i)$.

2.3 Leakage–safe geometric projection

A naive approach to recording shot events would be to store the final ball location, either at the goalkeeper intervention or at the goal line. This practice, however, introduces *outcome leakage*. For instance, if the ball is caught outside the posts, the data immediately reveals that no goal was scored, whereas if the ball crosses the line at $x = 0$, the outcome is implicitly known to be a goal. To avoid such bias, we impose a common reference frame: **all shots are projected to the vertical goal plane $x = 0$ using only strike-time information.**

We fix a pitch-based coordinate system with x pointing towards goal, y lateral, and z vertical. At strike time, the ball is located at ($\texttt{startX}, \texttt{startY}$) with $\texttt{startX} > 0$, and its horizontal trajectory is $(\Delta x, \Delta y)$, where shots directed toward goal satisfy $\Delta x < 0$. Assuming the ball follows a straight trajectory from the strike point, the parametric equation of motion in the horizontal plane is

$$(x, y) = \big(\texttt{startX}, \texttt{startY}\big) + \alpha\,(\Delta x, \Delta y).$$

[1] FootoVision provides state-of-the-art video data collection and a customizable analysis platform for professional clubs and federations.

The intersection with the goal plane $x = 0$ occurs for

$$\alpha^* = -\frac{\texttt{startX}}{\Delta x},$$

which yields the projected lateral intercept

$$\widehat{\texttt{endY}} \;=\; \texttt{startY} - \frac{\texttt{startX}\,\Delta y}{\Delta x}. \tag{2}$$

An analogous calculation with the vertical component of the velocity provides $\widehat{\texttt{endZ}}$.

The pair $(\widehat{\texttt{endY}}, \widehat{\texttt{endZ}})$ therefore describes where the ball *would intersect the plane $x = 0$ if unobstructed*, using only information available at strike time. This construction guarantees three key properties:

1. projections are defined for all attempts, regardless of whether the shot is saved, blocked, or scored;
2. no post-outcome information (such as save location or actual touch point) contaminates the model;
3. every attempt is represented relative to the same reference plane, making finishing patterns directly comparable.

Although the straight-line assumption ignores spin, curve, or dip, it provides a leakage-safe and computationally tractable approximation.

2.4 Model: GA^{2}M learned by Explainable Boosting

We model the log-odds function $F : \mathcal{X} \to \mathbb{R}$ of the conditional post–shot success probability

$$p(x) \;=\; \mathbb{P}(Y = 1 \,|\, X = x,\, T = 1)$$

using a generalized additive model with selected pairwise interactions (GA^{2}M). Formally,

$$F(x) \;=\; \beta_0 \;+\; \sum_{j=1}^{M} f_j(x_j) \;+\; \sum_{j<k} f_{jk}(x_j, x_k), \qquad p(x) \;=\; \sigma\big(F(x)\big), \tag{3}$$

where $\sigma(t) = (1 + e^{-t})^{-1}$ denotes the logistic link. The univariate functions f_j capture main effects, while the bivariate functions f_{jk} describe interactions within a sparse, domain-driven set $\mathcal{I}$.

Identifiability. To ensure uniqueness of the decomposition, we impose mean-centering with respect to the training distribution:

$$\mathbb{E}[f_j(X_j)] = 0, \quad \mathbb{E}[f_{jk}(X_j, X_k)] = 0, \qquad \forall j,\ (j,k) \in \mathcal{I}.$$

Under this convention, β_0 corresponds to the baseline log-odds

$$\beta_0 = \operatorname{logit}\big(\mathbb{P}(Y = 1 \,|\, T = 1)\big),$$

and each shape function f is interpreted as a deviation from that baseline.

Estimation. Let $\mathcal{F}_{\mathrm{GA^2M}}$ denote the function class in (3) where f_j and f_{jk} are restricted to be piecewise-constant over adaptively chosen bins. Given a training sample $\{(x_i, y_i)\}_{i=1}^n$ of n i.i.d. attempts with $T_i = 1$, estimation proceeds via regularized empirical risk minimization under logistic loss:

$$\widehat{F} \in \arg\min_{F \in \mathcal{F}_{\mathrm{GA^2M}}} \frac{1}{n} \sum_{i=1}^{n} \ell(F(x_i), y_i), \qquad \ell(F, y) = \log\big(1 + \exp(-(2y-1)F)\big).$$

Regularization is achieved implicitly through shrinkage, early stopping, and bagging. In practice, we rely on Explainable Boosting Machines (EBMs), which perform stagewise fitting of f_j and f_{jk} using shallow regression trees. Details of the training algorithm are deferred to Appendix A.

The interaction set $\mathcal{I}$ is deliberately sparse and guided by domain knowledge; for instance, interactions between lateral/vertical intercepts and shot speed, or between initial distance and lateral target, are included to capture key physical dependencies.

Interpretability: exact additive attributions and odds ratios. Recall that $p(x) = \mathbb{P}(Y = 1 \mid X = x, T = 1) = \sigma(F(x))$, where

$$F(x) \;=\; \beta_0 + \sum_j f_j(x_j) + \sum_{(j,k)\in\mathcal{I}} f_{jk}(x_j, x_k).$$

Because F is additive in low dimensions, the model admits an exact decomposition of contributions. In particular, changing an input configuration from x to x' modifies the goal *odds* multiplicatively:

$$\frac{p(x')/(1-p(x'))}{p(x)/(1-p(x))} \;=\; \exp\!\left(\sum_j \big[f_j(x'_j) - f_j(x_j)\big] \;+\; \sum_{(j,k)\in\mathcal{I}} \big[f_{jk}(x'_j, x'_k) - f_{jk}(x_j, x_k)\big]\right).$$

In other words, if we compare the probability of scoring for the observed attempt x to that of a counterfactual attempt x' (e.g. the same shot but struck faster, or aimed closer to the post), then the odds ratio depends *exclusively* on the difference in contributions of the shape functions f_j and f_{jk}. Thus both local explanations (for individual shots, via the terms $f_j(x_j)$ and $f_{jk}(x_j, x_k)$) and global patterns (the shapes themselves) are inherent to the model. Unlike post-hoc surrogate methods, interpretability is built directly into the functional form.

Reliability: calibration and its assessment. Under logistic loss, the population risk minimizer satisfies $F^\star(x) = \operatorname{logit} p^\star(x)$, so that $p^\star(x) = \sigma(F^\star(x))$ is, by definition, perfectly calibrated. In practice, EBMs approximate $F^\star$ while controlling variance through shallow learners, shrinkage, sparse interactions, and bagging, which empirically yields well-behaved probability estimates.

We evaluate calibration using two complementary tools. First, *reliability diagrams* provide a visual assessment of agreement between predicted probabilities and observed frequencies. Second, we report the *Expected Calibration Error* (ECE),

$$\text{ECE} \;:=\; \sum_{k=1}^{K} \frac{|B_k|}{n}\, \big|\, \overline{p}_{B_k} - \overline{y}_{B_k} \,\big|,$$

where $(B_k)_{k=1}^K$ denotes a partition of $[0,1]$, $\overline{p}_{B_k}$ is the average predicted probability in bin B_k, and $\overline{y}_{B_k}$ the corresponding empirical goal rate (see Appendix B for more details).

As a conservation check, we also compare aggregated predictions to actual outcomes on the test split:

$$\sum_{i\in\text{test}} \widehat{p}(x_i) \;\approx\; \sum_{i\in\text{test}} y_i,$$

and report both absolute and relative discrepancies. In our experiments (Section 2.6), calibration proves strong, and no post-hoc correction such as isotonic regression or Platt scaling is required. In addition to the Expected Calibration Error, we also report the *Brier score loss*, defined as

$$\text{Brier} := \frac{1}{n}\sum_{i=1}^{n} \big(\widehat{p}(x_i) - y_i\big)^2.$$

The Brier score measures the mean squared distance between predicted probabilities and observed outcomes, and thus captures both calibration and sharpness of predictions in a single metric. Lower values indicate more reliable probabilistic forecasts. In our experiments, EPS achieved the best Brier score among all models (Table 1), reinforcing its ability to deliver accurate and well-calibrated probability estimates.

2.5 From shot probabilities to player indices

To assess player performance, EPS is aggregated differently depending on the role. For attackers, we assign a value of EPS_i to on–target attempts and *zero* to off–target attempts:

$$\widetilde{\text{EPS}}_i := \begin{cases} \text{EPS}_i, & t_i = 1, \\ 0, & t_i = 0, \end{cases}$$

so that every attempt is included in the player's record. In this way, shots that miss the target contribute a post–shot value of 0, while only on–target attempts add positive execution quality.

For goalkeepers, evaluation relies by summing EPS across all on–target shots:

$$\mathrm{EPS}^{\mathrm{GK}} = \sum_{i:t_i=1} \mathrm{EPS}_i.$$

This quantity represents the reference number of goals an average goalkeeper would be expected to concede given the same sequence of attempts, and therefore provides a natural benchmark for comparing actual performance. EPS provides a calibrated measure of execution quality at the shot level, which can be aggregated into interpretable player indices for both goalkeepers and attackers.

Goalkeepers. For a goalkeeper g facing a set of shots S_g with observed outcomes $\{y_i\}$, we define the *Goalkeeper Rating* as

$$\mathrm{GKR}(g) \;:=\; \frac{\sum_{i\in S_g} \mathrm{EPS}_i \;-\; \sum_{i\in S_g} y_i}{|S_g|}.$$

This index measures the average goals prevented relative to an EPS baseline: it is positive when the goalkeeper concedes fewer goals than predicted, and negative otherwise. In addition, we report the absolute goal differential $\sum_{i\in S_g} \mathrm{EPS}_i - \sum_{i\in S_g} y_i$, which provides a more intuitive measure of the total swing in match outcomes.

Attackers. Let $\mathcal{A}_p$ denote the set of attempts for player p, and let $xG(i)$ denote a standard pre–shot expected-goals model. We introduce two complementary *Striker Ratios*. The first index

$$\mathrm{STR}^{(1)}(p) \;:=\; \frac{\sum_{i\in\mathcal{A}_p} \widetilde{\mathrm{EPS}}_i \;-\; \sum_{i\in\mathcal{A}_p} xG(i)}{|\mathcal{A}_p|}$$

captures the average upgrade (or downgrade) from chance creation to shot execution. The second,

$$\mathrm{STR}^{(2)}(p) \;:=\; \frac{\sum_{i\in\mathcal{A}_p} \max\{\widetilde{\mathrm{EPS}}_i, y_i\} \;-\; \sum_{i\in\mathcal{A}_p} xG(i)}{|\mathcal{A}_p|},$$

ensures that successful finishes are never undervalued, even when the predicted execution probability is low.

Both indices derive directly from the calibrated glass-box model and therefore remain fully interpretable: for any player, deviations can be traced to concrete execution patterns such as shot placement, vertical height, or ball speed.

2.6 Model Performance: Discrimination and Calibration

Descriptive statistics (Figure C.3) confirm, for example, that the empirical distributions of `startY` and `endY` are symmetric around zero, suggesting no lateral bias relative to the goal center. We benchmark EPS against standard classifiers trained on the same stratified split and feature set. *Logistic regression* provides a linear-in-log-odds baseline. *Support Vector Machines (RBF kernel)* add a flexible margin and typically improve ranking despite being threshold-agnostic at training time. *Random Forests* and *XGBoost* represent tree ensembles that partition the covariate space into axis-aligned regions, a natural fit to tabular, spatial features. Our model also relies on trees but learns low-depth boosted stumps on adaptively binned features, yielding glass-box main effects and a small set of domain-guided pairwise interactions. This preserves interpretability while typically producing better calibration. For comparability, hyperparameters are kept close to library defaults, ensuring that the pipeline can be retrained quickly and consistently across leagues and seasons without expert retuning.

Table 1: Classification performance on the test set. Lower is better for ECE and Brier. *Prec(1)* = precision on the positive class (goals). ECE with $K = 20$ equal-width bins.

Model	AUC	Prec(1)	Recall(1)	Acc.	F1	ECE	Brier
Logistic Reg.	0.7335	0.6358	0.5876	0.7393	0.3615	0.0283	0.1701
Random Forest	0.8975	0.7758	0.6640	0.8397	0.6983	0.0377	0.1110
XGBoost	0.9103	0.7724	**0.7023**	0.8540	**0.7395**	0.0241	0.1056
SVM (RBF)	0.8855	0.7629	0.6488	0.8311	0.6793	0.0206	0.1191
Neural Net	0.8755	0.7529	0.6432	0.8291	0.6789	0.0187	0.1211
EBM (no inter.)	0.8918	0.7878	0.6142	0.8389	0.6903	0.0200	0.1040
EBM (EPS)	**0.9150**	**0.7992**	0.6721	**0.8559**	0.7331	**0.0173**	**0.1026**

Tree-based learners perform strongly, reflecting the suitability of axis-aligned partitions to the discretized geometry of football shots. The EBM underpinning EPS attains the highest AUC (0.9150), competitive accuracy (0.8559), the lowest calibration error (0.0173), *and the best Brier score* (0.1026), confirming both discrimination and probabilistic accuracy. XGBoost achieves the best F1 score but with weaker calibration than EBM. The ablation without interactions (pure GAM) still performs competitively (AUC 0.892, F1 0.690, Brier 0.1040), but falls short of the full GA^{2}M. This confirms that a sparse set of domain-guided pairwise terms contributes materially to ranking and recall, while preserving the additive glass-box structure.

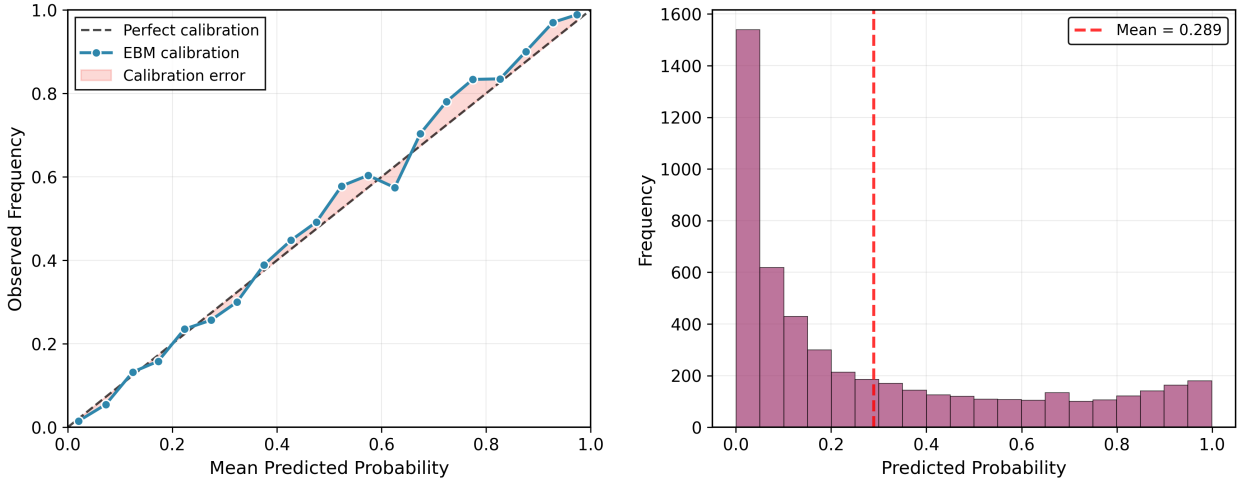


Figure 1: Calibration plot (left) and predicted probability histogram (right). The dashed diagonal in the left panel indicates perfect calibration. The right panel shows the distribution of predicted probabilities, with the red dashed line marking the proportion of goals in the test set.

At aggregate level, predicted and realized goals align closely (1693 vs. 1677, gap $< 1\%$). Replications over stratified seeds yield similarly low ECE and stable aggregates, supporting the conclusion that EPS probabilities are well calibrated *in distribution* (for details see Appendix D).

Finally, note that the neural-network baseline lags behind in this structured, tabular regime: with limited sample sizes per covariate configuration and sharp spatial discontinuities, shallow boosted trees over adaptive bins capture decision boundaries more effectively than generic deep architectures. A similar argument explains the weakness of logistic regression: a single global linear logit cannot reproduce the strong nonlinearities and interactions induced by spatial "grid" effects (e.g., far-post vs. near-post bands, speed–height synergies). Tree-based learners—including EBMs—natively tile the pitch and kinematic space into interpretable cells and low-dimensional shapes, matching the piecewise nature of football shot difficulty and yielding both higher discrimination and markedly better calibration.

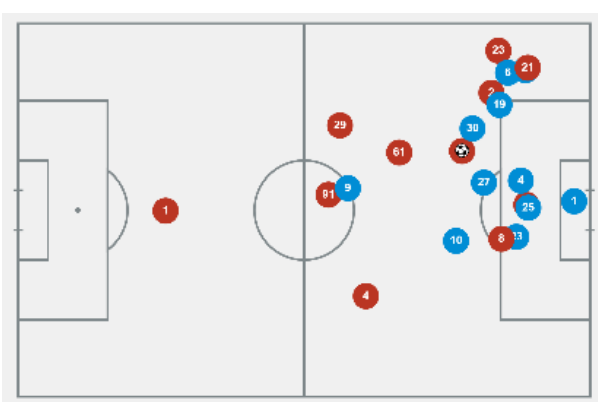
Implementation details. For reproducibility, we report the key hyperparameters used in all EBM runs. We relied on the library defaults, with the most relevant values being: learning rate $\eta = 0.01$, maximum boosting rounds = 25 000, maximum bins per feature = 1024, outer bagging folds = 14, interaction budget = 0.9 (fraction of features eligible for interactions), maximum leaves per tree = 3 (stumps), and validation split = 0.15. Early stopping was applied with a patience of 50 rounds and tolerance 10^{-5}. Additional defaults include no inner bagging, minimum samples per leaf = 2, and ℓ_2 regularization set to zero. This configuration reflects the intended design of EBMs: shallow learners, aggressive bagging, and shrinkage for variance control, combined with a sparse interaction search to preserve interpretability.

2.7 Interpretation: global and local decompositions of execution patterns

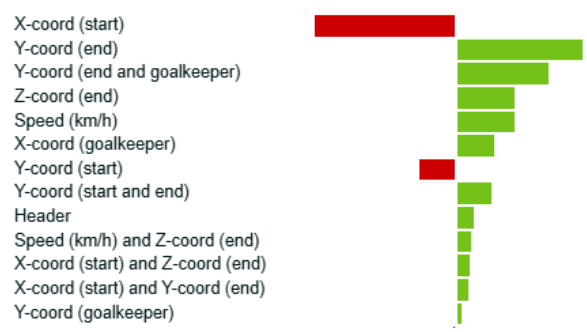
Global drivers and local reversals. Global importances (Figure E.4) confirm that the lateral intercept $\widehat{\texttt{endY}}$ is the dominant driver: moving the projected crossing point toward either post increases scoring probability. Geometric descriptors of the approach—`startX` (distance) and `startY` (lateral launch)—follow closely, alongside `speedKmh`, whose effect is amplified when combined with precise placement or height trajectories. Goalkeeper pose also contributes: shots crossing away from the keeper's column (`gkY`) or struck while the keeper is deeper (`gkX`) are favored. The `header` indicator has a modest main effect but interacts meaningfully with `endZ` and `startX`.

Importances are computed as the mean *absolute* contribution of each learned shape over the evaluation set; they quantify *how much* a term moves predictions on average, not *in which direction* or under which contexts. Consequently, the global ordering may reverse on a given play.

In the attempt of Fig. 2, the shot originates from $(\texttt{startX}, \texttt{startY}) = (29.9, 16.7)$ with speed 88.3 km/h, vertical intercept $\widehat{\texttt{endZ}} = 2.08$ m, and lateral intercept $\widehat{\texttt{endY}} = 2.28$ m. These values correspond to a fast strike aimed high and wide toward the right post. Locally, near–post placement ($\widehat{\texttt{endY}} = 2.28$) and high speed (88.3 km/h) supply the largest positive increments through the univariate shapes, while the long distance (29.9 m) contributes negatively. Because the launch is also wide on the opposite side, the interaction surface $f_{(\texttt{startY}, \widehat{\texttt{endY}})}$ adds a further boost, consistent with a curled trajectory returning toward the corner. In other contexts—for example, a central $\widehat{\texttt{endY}}$ combined with large `gkX` (keeper far off his line)—keeper–position terms could dominate locally despite ranking lower globally. This variability of local orderings is expected from the geometry of the play and the goalkeeper's position.



(a) 2D context (attack left → right), before the shot



(b) Exact additive decomposition for the attempt ($p = 0.70$).

Figure 2: Side–by–side view of context and local additive explanation for a single shot. The local bar plot shows the signed contributions of main effects and selected interactions.

2.8 Goalkeeper and Attacker Evaluation

With EPS defined at shot level and its calibration verified, we now turn to player evaluation.

Goalkeepers. For a goalkeeper, cumulative danger faced is contrasted with realized concessions through the GKR index, normalized by the number of on-target shots to ensure comparability across unequal minutes. Table 2 illustrates results for the first half of the 2024/25 EPL. The ranking aligns with domain priors: the top tier features established shot-stoppers, while the lower tier contains keepers whose concessions exceed the expected level. Ederson, Alisson, and Vicario appear near the top, reflecting a positive per-shot differential between expected and realized goals conceded, whereas Ramsdale and Palmer fall at the bottom, consistent with underperformance relative to shot difficulty.

Table 2: Example GKR ranking (EPL, 2024/25).

Goalkeeper	$\sum$EPS	$\sum$Goals	Shots on Target	**GKR**
Ederson (MCI)	16.72	15	57	+0.03
Alisson (LIV)	23.13	25	75	+0.03
Vicario (TOT)	30.55	29	86	+0.02
...	...	...	...	...
Ramsdale (SOU)	44.72	53	144	−0.06
Palmer (LEI)	18.49	23	39	−0.08

Attackers. For attackers, we quantify the ability to translate pre-shot chance quality into post-shot danger and, ultimately, into goals using the $\mathrm{STR}^{(1)}$ and $\mathrm{STR}^{(2)}$ indices (Section 2.5). $\mathrm{STR}^{(1)}$ captures the average upgrade from xG to EPS across a player's attempts. $\mathrm{STR}^{(2)}$ substitutes EPS with $\max\{\mathrm{EPS}, y\}$ so that rare but decisive low-probability conversions are credited at face value.

The two indices reveal complementary finishing profiles. Table 3 highlights players such as Welbeck, Wilson, and Schade, who systematically "improve" their chances at execution and rank highly in $\mathrm{STR}^{(1)}$. Conversely, Table 4 shows Wood, Larsen, and Bowen excelling in $\mathrm{STR}^{(2)}$, where realized scoring output drives the ratio upward. At the other end, major creators like De Bruyne and Fernandes register slightly negative values in $\mathrm{STR}^{(1)}$, indicating that their finishing this season has not upgraded pre-shot chance quality to the same extent as in past campaigns—consistent with a dip in finishing efficiency rather than in chance involvement.

Table 3: Example $\mathrm{STR}^{(1)}$ (EPL, 2024/25).

Player	$\sum$EPS	$\sum$xG	Goals	Shots	$\mathbf{STR}^{(1)}$
Welbeck (BRI)	8.90	4.93	8	39	+0.10
Wilson (FUL)	6.99	3.94	6	35	+0.09
Schade (BRE)	9.61	6.21	10	43	+0.08
...	...	...	...	...	...
Fernandes (MUN)	3.33	4.36	2	28	−0.01
De Bruyne (MCI)	0.96	1.75	1	17	−0.02

Table 4: Example $\mathrm{STR}^{(2)}$ (EPL, 2024/25).

Player	$\sum \max(\mathrm{EPS}, y)$	$\sum$xG	Goals	Shots	$\mathbf{STR}^{(2)}$
Wood (NFO)	17.40	1.90	15	50	+0.17
Larsen (WOL)	13.69	7.23	10	39	+0.16
Bowen (WHU)	13.30	4.84	12	54	+0.15
...	...	...	...	...	...
Calvert-Lewin (EVE)	4.72	4.20	3	38	0.01
Savinho (MCI)	3.51	3.07	1	36	0.01

Taken together, these two perspectives isolate complementary facets of finishing. $\mathrm{STR}^{(1)}$ emphasizes systematic, repeatable execution gains over the pre-shot baseline, while $\mathrm{STR}^{(2)}$ reflects realized impact on the scoreboard. Crucially, both derive from a calibrated, glass-box model: analysts can move seamlessly from player aggregates back to the learned shapes f_j and f_{jk} to diagnose concrete execution patterns—placement, height, or speed—that underpin each player's profile.

3 Cross-League Transfer Projection (TRAM)

Beyond within-league evaluation, EPS also enables cross-league transfer analysis. The guiding idea is instance-level analogy: for each shot taken in a *source* league L_1 (e.g., Portugal), we retrieve a small set of *most similar* shots observed in a *target* league L_2 (e.g., England), compare the corresponding EPS values, and aggregate at the player level. TRAM thus complements GKR and STR: whereas GKR captures shot-stopping, and STR captures finishing efficiency within a league, TRAM quantifies how finishing execution is expected to translate across competitions.

3.1 Methodology

For a source shot i with descriptor

$$z_i \;=\; \big(\texttt{expectedGoal},\, \texttt{startX},\, \texttt{startY},\, \widehat{\texttt{endY}},\, \widehat{\texttt{endZ}},\, \texttt{speedKmh},\, \texttt{header}\big),$$

we compute its k nearest neighbours $\mathcal{N}_k(i) \subset L_2$ using Euclidean distance in standardized feature space (each coordinate centered and scaled on $L_1 \cup L_2$). Unless specified otherwise, $k = 5$ offers a good bias–variance compromise.

The shot-level differential is defined as

$$\Delta(i;\, L_1 \to L_2) \;:=\; \widetilde{\mathrm{EPS}}_{L_1}(x_i) \;-\; \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} \widetilde{\mathrm{EPS}}_{L_2}(x_j),$$

i.e. the gap between observed execution in L_1 and the typical execution baseline in L_2. For a player s with attempts $\mathcal{A}_s$ in L_1, the *Transfer Metric* is

$$\mathrm{TRAM}(s;\, L_1 \to L_2) \;:=\; \frac{1}{|\mathcal{A}_s|} \sum_{i \in \mathcal{A}_s} \Delta(i;\, L_1 \to L_2). \tag{4}$$

A positive TRAM indicates that, on like-for-like shots, a player's execution in L_1 exceeds the baseline observed in L_2, with units interpretable as goals per shot (e.g., $+0.05 \approx 5$ extra goals per 100 attempts).

3.2 Results: Portugal → England

Our analysis processed 5030 shots from Portuguese players, with 52 players meeting the minimum threshold of 25 attempts. The aggregate TRAM of -0.027 indicates that Portuguese finishing translates slightly below EPL standards, though dispersion is high (s.d. ≈ 0.123). Table 5 reports the strongest positive and negative differentials.

Table 5: TRAM projection from Portuguese Primeira Liga to English Premier League.

Player	$\sum \mathrm{EPS}_{\mathrm{PT}}$	$\sum \mathrm{xG}_{\mathrm{PT}}$	Goals	Shots	Baseline $\mathrm{EPS}^{(k)}_{\mathrm{ENG}}$	**TRAM**
Gyökeres (SPO)	28.92	9.94	26	90	21.33	+0.084
Alan (GUI)	5.34	2.41	4	30	3.21	+0.071
Horta (BRA)	11.52	4.47	11	39	8.86	+0.068
...	...	...	...	...	...	...
Nuno Santos (GUI)	4.97	3.09	3	37	9.02	−0.109
Pohlmann (RIO)	2.23	2.12	2	31	6.18	−0.128

The distribution shows clear heterogeneity. **Gyökeres** displays the strongest positive differential (+0.084): on matched contexts, his execution in Portugal outperforms the EPL baseline by roughly 8.4 goals per 100 shots, supported by 90 attempts. At the other end, **Pohlmann** (−0.128) and **Nuno Santos** (−0.109) fall well below, suggesting their finishing would translate to fewer goals per 100 shots relative to comparable EPL contexts.

Overall, TRAM identifies both league-level shifts and standout individuals. Portuguese finishing appears, on average, slightly below EPL execution standards, but positive outliers highlight players whose finishing skill is likely robust to competition shifts.

3.3 Statistical considerations and biases

Sample sizes range from 25 to 90 shots per player, which yields informative but not definitive precision. Estimates at the individual level remain subject to sampling variability, and confidence intervals—for instance via bootstrap—are desirable when reporting player-specific values.

Our k-NN matching controls for strike location, projected endpoint, speed, body part, and pre-shot danger, but unmeasured factors such as defensive pressure, goalkeeper quality mix, and game-state effects may still influence comparisons. Moreover, by conditioning on on-target attempts, EPS focuses on execution once the ball is framed; off-target misses are not captured, which could bias evaluations of players with systematically poor shot selection.

Finally, transfer projections assume comparable shot distributions across leagues. Differences in tactical styles (e.g., long-range attempts vs. cutbacks) and goalkeeper pools can alter the effective baseline. As such, EPS-based indices should be interpreted with uncertainty margins and contextual awareness of league and match dynamics.

Conclusion

We introduced **EPS**, a calibrated and intrinsically interpretable post-shot metric based on Explainable Boosting Machines. EPS achieves state-of-the-art discrimination (AUC 0.915) while maintaining strong calibration (ECE ≈ 0.017), and its glass-box decomposition yields transparent, human-readable effects. Building on these foundations, we defined recruitment-oriented indices: GKR for goalkeepers, $\mathrm{STR}^{(1)}/\mathrm{STR}^{(2)}$ for attackers, and TRAM for cross-league transfer projection. Together, these metrics convert on-target execution into decision-ready aggregates.

Methodologically, EPS demonstrates that interpretable additive models can rival black-box ensembles in predictive power while offering calibrated probabilities and auditable explanations. Practically, the framework provides clubs and federations with tools to assess finishing quality, goalkeeper shot-stopping, and transferability across competitions.

Future work will integrate richer ball- and goalkeeper-tracking data, more refined trajectory models, and formal uncertainty quantification for TRAM and counterfactuals. Beyond individual evaluation, extensions to squad-level finishing environments and real-time applications may further enhance tactical and recruitment analysis. In sum, EPS bridges the gap between statistical rigor and practical interpretability, offering a transparent foundation for modern football analytics.

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Appendix

A EBM Training Algorithm for Binary Classification

Below is a generic outline for training an EBM for binary classification using gradient boosting of shallow trees.

Algorithm 1 EBM Training for Classification

Require: $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{0,1\}$
η : learning rate (shrinkage parameter)
M : maximum number of boosting iterations
InteractionPairs : set of feature pairs to model interactively

Initialize :

$$F^{(0)}(\mathbf{x}) \leftarrow \log\Big(\frac{\bar{y}}{1-\bar{y}}\Big), \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

▷ Initial log-odds as global average

for $m = 1$ to M **do**
  $r_i^{(m)} \leftarrow y_i - p_i^{(m-1)}, \quad$ where $p_i^{(m-1)} = \frac{1}{1+e^{-F^{(m-1)}(\mathbf{x}_i)}}$ ▷ Compute pseudo-residuals
  for each *univariate* feature $\ell \in \{1, \ldots, d\}$ **do**
    Fit a depth-1 tree (stump) to $\{r_i^{(m)}\}$ using $\mathbf{x}_i^\ell$
    Let the fitted tree for feature ℓ be $h_\ell^{(m)}(x_\ell)$
  end for
  for each *pairwise* $(j,k) \in$ InteractionPairs **do**
    Fit a depth-1 tree to $\{r_i^{(m)}\}$ using $(\mathbf{x}_i^j, \mathbf{x}_i^k)$
    Let the fitted tree be $h_{jk}^{(m)}(x_j, x_k)$
  end for
  $F^{(m)}(\mathbf{x}) \leftarrow F^{(m-1)}(\mathbf{x}) + \eta \sum_{\ell=1}^{d} h_\ell^{(m)}(x_\ell) + \eta \sum_{(j,k)\in \text{InteractionPairs}} h_{jk}^{(m)}(x_j, x_k)$
  ▷ Update model with shrinkage η
  check stopping criterion (*e.g.*, if validation loss improves minimally)
end for
Return $F^{(M)}(\mathbf{x})$

Remarks :

- The user must specify which pairs of features to include in InteractionPairs. Some implementations automatically detect pairs that reduce the loss most significantly.
- Each stump is typically just a single split on that feature (or pair).
- The final $\hat{p}_i$ is obtained via logistic mapping : $\sigma\big(F^{(M)}(\mathbf{x}_i)\big)$.

B Algorithm for Evaluating Calibration

Algorithm 2 Procedure for Computing Calibration Metrics (e.g., ECE)

Require: $\{\hat{p}_i, y_i\}_{i=1}^n$, where $\hat{p}_i \in [0,1]$ is the predicted probability of a positive outcome for instance i, and $y_i \in \{0,1\}$ is the observed outcome.
K : number of bins (e.g., 10 bins for deciles).

Step 1 : Discretization
Divide the interval $[0,1]$ into K bins of equal width :

$$\left[0,\ \tfrac{1}{K}\right),\ \left[\tfrac{1}{K},\ \tfrac{2}{K}\right),\ \ldots,\ \left[\tfrac{K-1}{K},\ 1\right].$$

▷ Larger K gives finer granularity but may increase variance in each bin.

Step 2 : Grouping
for $i = 1$ to n **do**
  Identify the bin index k such that

$$\frac{k-1}{K} \ \leq\ \hat{p}_i \ <\ \frac{k}{K}.$$

  Assign $\hat{p}_i$ to bin B_k.
end for
Step 3 : Calculate Statistics in Each Bin
for $k = 1$ to K **do**
  Let $n_k = |B_k|$ be the number of predictions in bin B_k.
  Compute the mean predicted probability in bin B_k :

$$\hat{p}_{\text{mean},k} \ =\ \frac{1}{n_k} \sum_{i \in B_k} \hat{p}_i.$$

  Compute the empirical frequency (observed outcome rate) :

$$\hat{y}_{\text{emp},k} \ =\ \frac{1}{n_k} \sum_{i \in B_k} y_i.$$

end for
Step 4 : Compute Expected Calibration Error (ECE)
Let $n = \sum_{k=1}^{K} n_k$. Then

$$\text{ECE} \ =\ \sum_{k=1}^{K} \frac{n_k}{n} \left| \hat{p}_{\text{mean},k} \ -\ \hat{y}_{\text{emp},k} \right|.$$

▷ Lower ECE values imply better calibration.

Return : ECE (and intermediate statistics if needed).

C Descriptive statistics

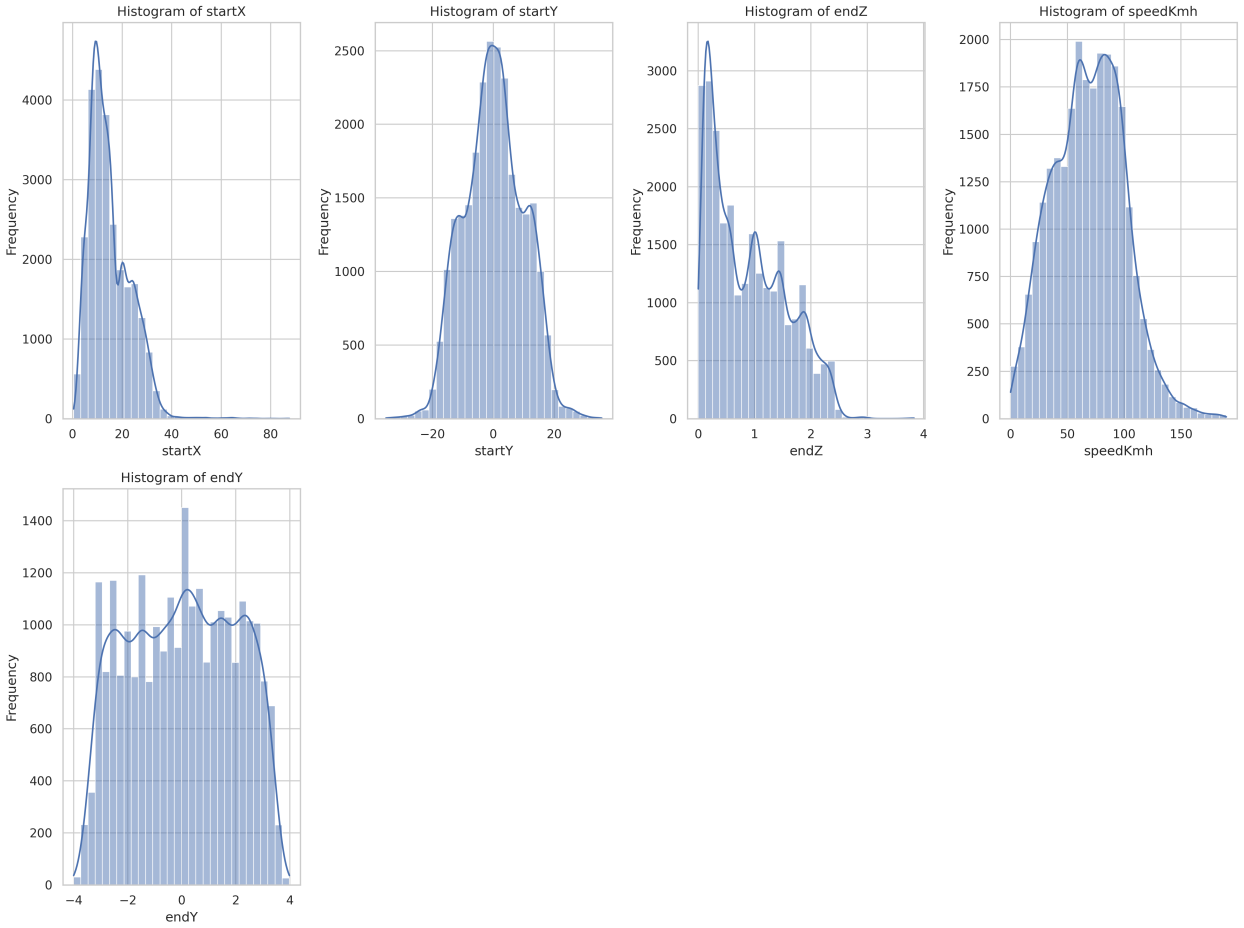

Figure C.3: Empirical distributions of key shot descriptors. Distances (`startX`) concentrate between 5 and 25m, reflecting the prevalence of penalty-box attempts. Lateral launch positions (`startY`) are approximately symmetric around zero, with few extreme wide shots. The vertical intercept (`endZ`) is bimodal, with a large mass near ground level and secondary peaks around 1–2m. Lateral intercepts (`endY`) are evenly spread across the goal width, confirming no systematic side bias. Finally, ball speed (`speedKmh`) follows a bell-shaped distribution centered around 70–100 km/h, with tails corresponding to very slow and very powerful strikes.

D Calibration results

Table D.6: Bin-wise calibration results with $K = 20$ equal-width bins (test set $n = 5109$). ECE = 0.0173. Lower is better.

Bin	**Predicted Mean**	**Empirical Freq.**	**Count**	|Error|
1	0.0205	0.0150	1531	0.0055
2	0.0728	0.0547	622	0.0182
3	0.1241	0.1323	431	0.0081
4	0.1729	0.1582	297	0.0147
5	0.2231	0.2350	217	0.0120
6	0.2739	0.2567	187	0.0172
7	0.3236	0.3000	170	0.0236
8	0.3750	0.3889	144	0.0139
9	0.4270	0.4480	125	0.0210
10	0.4753	0.4917	120	0.0163
11	0.5236	0.5780	109	0.0544
12	0.5744	0.6036	111	0.0292
13	0.6251	0.5743	101	0.0509
14	0.6743	0.7037	135	0.0294
15	0.7239	0.7800	100	0.0561
16	0.7743	0.8333	108	0.0591
17	0.8268	0.8347	121	0.0079
18	0.8757	0.9000	140	0.0243
19	0.9274	0.9699	166	0.0425
20	0.9738	0.9885	174	0.0147

The Expected Calibration Error is ECE = 0.0173 across $K = 20$ bins, confirming good reliability. Aggregate quantities also align: the model predicts 1477.7 goals versus 1493 observed (relative gap $\approx 1.0\%$). The mean predicted probability (0.2892) closely matches the observed goal rate (0.2922), with a prediction–realization gap of -0.0030. Overall, these results confirm that EPS is well calibrated *in distribution*.

E Global Feature Importances

Figure E.4 shows the global feature contributions learned by the EBM. These importances are computed as mean absolute contributions across the evaluation set, and quantify how much each term moves predictions on average.

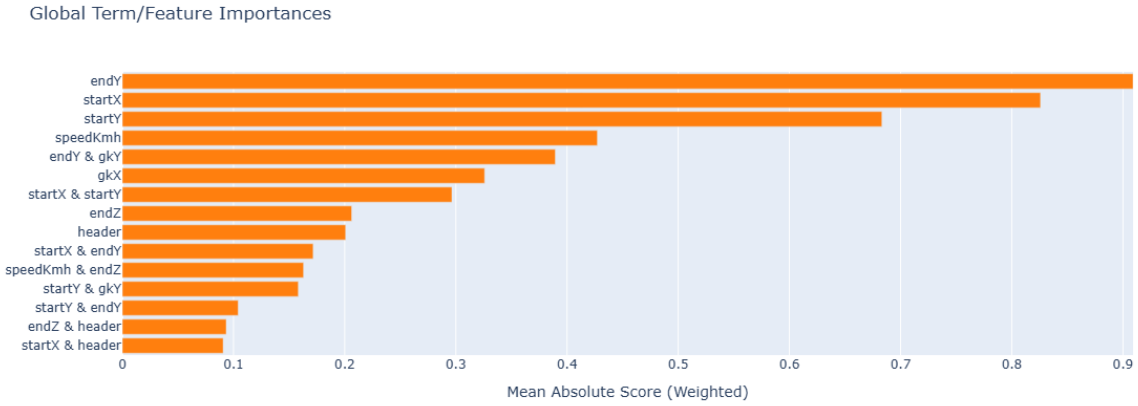

Figure E.4: Global feature importances from the EBM.